Welcome to Myrodia

This is the main working document; alongside this is a risk assessment, a presentation and our Gantt Chart for time management.

We hope you enjoy the culmination of our efforts!

- Rami, Veer, Yvan and Sai

What is Myrodia?

The Greek were renowned for their use of fragrance, often offering it to their Gods. In light of this, we decided to link the future of scent back to its humble pasts with the Greek name for smell: Myrodia.

Myrodia is a highly-specific device that brings a 4D experience to the homes in a both compact and affordable fashion. It brings together chemical, physical and technological engineering in a beautiful harmony to produce great outcomes.

The basis is a machine that exists alongside a computer / console, that is able to replicate senses to create greater immersion and an overall heightened experience. Placed on a desk, we envision it to send smells in the direction of the consumer, in a fast-paced, short-term manner.

The Team

Rami Al-Kheralla - Team Lead - involved in all aspects of the project, but largely responsible for putting together the different aspects in a project involving so many disciplines. Also, organising meetings, checking up on progress and time keeping.

Yvan Fejio-Tiofack - Tech lead - Designed the electronics of the project, as well as linking them to the game. Completed research on the best ways to do this, comparing chipsets and transfer means.

Veer Parekh - Physical operation lead - involved in creating the game and the different environments corresponding to the smells we have purchased. Also involved in the science behind everything and making sure the Myrodia machine and the game work together.

Sai Kaushal Chilamakuru - Chemical lead - mainly involved in the scents; procuring the different scents for different biomes in our game, finalising the concentration of scent used for the machine, testing how each scent will affect the user and the environment around.

What is Myrodia? - Our aim - 1.1

The Greek were renowned for their use of fragrance, often offering it to their Gods. In light of this, we decided to link the future of scent back to its humble pasts with the Greek name for smell: Myrodia.

Myrodia is a highly-specific device that proves the concept of bringing a 4D experience to the homes in a both compact and affordable fashion. It brings together chemical, physical and technological engineering in a beautiful harmony to produce great outcomes.

The basis is a machine that exists alongside a computer / console, that is able to replicate senses to create greater immersion and an overall heightened media experience. Placed on a desk, we envision it to send mists, gusts and notably smell in the direction of the consumer. To show that this is possible, our presented device should be able to work with one game, which we have designed. This presents that with more work, a final product is possible.

To achieve this, the following objectives must be met:

- Design and develop a game to display the technology; proving the concept. This must be created in a way that it is linked to the device. Veer / Yvan

- Develop a system for smell dispensation as well as development of the smells themselves - ensuring necessary health and safety standards are met. Rami / Sai

- Design and development of the wind (vortex) and mist machines, and linkage to the smell to create a system that works simultaneously. Veer / Yvan

- Rigorous testing and improvement to ensure a high quality final product, leaving the prototype phase and entering the final stage of completion. Rami / Sai

Major meeting log; (approximately) every month

For our complete time organisation; please refer to the Gantt chart excel file attached alongside this document.

Meeting 1 - 07/11/25
Attendees : All

- Project confirmed
- Roles given
- Preliminary research begins
- Product road agreed, starting with prototype 1

Meeting 2 - 19/11/25
Attendees : All

- Name ideas : Myrodia, Keller, Tapputi
- Name chosen in the end : Myrodia (mee - ro - thee - AH)

Meeting 3 - 20/11/25
Attendees: All

- Yvan says we try to project smells to be associated with memories
- Rami wants to still test if it will work with blind-deaf people
- Veer states we make a vr game that will use the smell project and be in the game
- Put places with familiar smells in the game

Meeting 4 - 30/11/25
Attendees: Rami, Yvan and Veer (Sai unavailable)

- Hard bit is connecting it to a raspberry pi
- First make it a 3d game and then expand to vr afterwards
- Machine that already exists is three thousand pounds - we will make it much cheaper

Meeting 5 - 08/12/25
Attendees: Rami, Veer and Sai (Yvan unavailable)

- Veer says we don’t need moving parts - have loads of humidifiers that will direct the gases immediately instead of mixing liquids and then humidifying
- Nebuliser uses vibrations to make into a gas.
- Expand from only smell - wind, water, environment stuff

Meeting 6 - 15/01/26
Attendees: All

- Only coding required which we expect is C sharp with if statements
- More control from the raspberry pie than the computer.
- Veer and Yvan will start looking at the game
- Sai and Rami will start to research the chemistry part

Meeting 7 - 2/02/26
Attendees: Yvan, Sai and Veer (Rami unavailable)

- Place ideas : Swimming pool, Pizzaria, Garden / Outdoors, Picnic / Kitchen, Fart (potentially..), animal enclosure, lavender smell, tour of a random location or area, bakery.
- Use functions in a raspberry pi instead of lots of code.
- Make a dictionary and function for the smells.

Meeting 8 - 13/04/26
Attendees: All

- Scents and diffusers have arrived; experimenting begins!
- The first prototype is seen in action; works well with water, however when combined with the scent, the dispersion is not as effective.
- Questions raised regarding the concentrations of scent to water for the most effective release of smell, where the scent remains strong.

Meeting 9 - 17/05/26

- Sai and Rami introduced new concentrations to the group, 40% scent, 60% water to the group, which going up in ratios of 10%, was the most effective balance of smell and dispersion.
- With the diffusers working, focus shifts to the game, Veer is assigned to the development of this. Yvan carefully assists, as he will focus on the connection between the game and the machine

Meeting 10 - 17/06/26
Attendees: All

- The game is designed, and played through on the laptop. It is as intended. Yvan now begins looking into connectivity.
- It is discussed that whilst initially a Raspberry Pi would be used, an Arduino now seems better suited due to the nature of the code.

Meeting 11 - 18/07/26
Attendees: All

- Yvan demonstrates the written code (see pasted further below).
- The game works! It triggers the smells, achieving our aim.
- The final task - finalising our document for submission.

Meetings with Steve, an electrical engineer with CREST experience

- Nature: On a ‘need’ basis; booked in via email.

We would like to begin this section with a huge thanks to Steve for his immense efforts in passing his insights to us. We really do appreciate it. Thank you very much and Mr Mohan, our Supervisor, for helping us get into and regulating our contact.

Steve’s information:
Email: steve@smythe.me.uk
Degree: Electronics and physics

Meeting 1- 25/02/2026
Attendance: All

General Crest Award Insight:
-30% pass: ensure that all of the 15 requirements are met
-Have all evidence for work done on project
-Look at previous projects that are similar
-11/15 required to pass assessment
Meetings will occur in 2 weeks (11/03/2026)

Required fix

Ensuring that the slides are correlated with the project	All
How will the scent in the air go away, due to potential scent mix occurring	Rami
How electronically will the system be compatible	Veer

Frequent reasons for the failure of projects:

Reasons for failure	Explanations	Fix	Responsibilities	Completed
Failure to document programming involved	-Not demonstrating coding	-Document all evidence of work through github etc, ensuring that it is visible to assessor	Yvan	✓
Failure to document the science involved	-Not demonstrating you know how the machine works	-Document the physics behind everything	Veer	✓
Not including items used	Unclear list of all items used making it of development process	Include clear list of all items used before combining the	Yvan	✓
Have clear proof of concept	-Unclear logic process to solution	-Documentation of all methods within electronics	Veer	✓
Failure to reflect on issues	-Functionality is not priority but rather reflection on issues faced and evaluation	-With each issue faced document all steps taken to overcome	Rami	✓
Assessment point 2.2 (Reflection)	Make sure to include all resources that are used, **do it as we go along**.	-use **MyBib** to list references with correct format -Ensure that all methods/steps taken are documented with sufficient detail to be able to replicate	Sai	✓
Not utilising the student profile form	-Not utilising student profile form to demonstrate skills used to the assessor	Documenting all skills used within the student profile	All (Sai can you find out more about student profile)	✓
Assessment point 2.1	-Not utilising material and people to support the project	Get in contact with those that can support the project	All	✓
Assessment point 1.3	Evaluation of aims and objectives not covering all potential methods	consider other approaches to meet aims and objectives (research,video,physical item)	Rami	✓
Assessment point 1.4	This criterion is about stating the plan for the project at a high-level with students reflecting on and justifying their choices.	Describe plan of project -Ensure to include reasons for why other approaches are not taken -make clear why current plan is used and advantages	Rami	✓
Devices	Variety of devices unconsidered	Consider devices used and pros and cons and the impact on the user	Yvan	✓

Meeting 1- 18/03/2026
Attendance: All

- Main objective : discuss the functionality and update on the project
- Main concern: how will we connect the raspberry pi and game (lan or wireless)
  ENSURE NOT TO LEAVE TILL END
- Could use items such as Amazon dash
- How will we get the scent machine get released leaning toward ultra sonic motor
- Issues addressed: 4 relays needed

General tips:
A 3D case will not be needed, it is a proof of purpose, rather than a for-market project. This could be an interesting goal for the future however.
Document all issues faced and how we learned

Moving forward:
Order products
Veer makes simple game
Yvan should test the game
Steve contact us in regards to motor

The Science behind Myrodia

Myrodia will be split into 2 parts: the game and the machine. The game will consist of a simple 3D game using Unity, that consists of 4 different environments and a character that the user is free to move in. This game will be precoded which means more time can be spent on the actual Myrodia machine and providing the smells. This game will have 4 different environments and the game will be able to detect which environment the character will be in using the coordinates of the game. These coordinates can then be transferred to the Myrodia machine so the corresponding smells can be released.

The Myrodia machine was initially going to be made with a raspberry pi but we decided to use an arduino instead. This is because an arduino had more compatible components which allowed us to release the smells with cheaper and easier to use components. We then had to connect the game with Myrodia, which would be done using 1 of 3 methods: A wired connection, using a local area network(LAN) or using bluetooth. We decided to use a wired connection because it was the easiest to do and wasn't too complicated to get working. The arduino will then be programmed in C++ and depending on the coordinates of the character in the game, a different smell will be produced by the Myrodia machine. Each environment corresponds to a different smell and each smell will have its own diffuser which can be turned on and off individually.

Wider purpose for project - 1.2

Whilst the direct purpose for the product is to make consuming entertainment more enjoyable at home, there are a number of secondary purposes.

We target this product at helping people who cannot enjoy the blessings of life as much as us fortunate enough to.

Firstly, those who are blind or deaf who grow to have an increased dependency on their other senses. They can learn to understand plots and narratives not from what is on the screen or what they hear, but from what they smell and feel. It allows entertainment directed to the mainstream to be enjoyed by all.

Secondly, those who may have illnesses such as those that cause a weak immune system. In this case, people are restrained to their homes, preventing them from experiencing the beauty of the world, through its senses which drive our emotions. Our device envisions a future where we collaborate with virtual reality to give these people a life with stimulation and joy.

Issues experienced and our solutions

Our project has developed in a number of ways from its infancy.
Initially, it was to be a smell-only device, targeted towards individuals who are blind and deaf, using smell as a third sense in identifying objects.

However, this idea whilst promising on paper, fell short, with smell being a less contributing factor in identification for those where touch is still an option.

Therefore, we moved forward, keeping the same premise - different objective. We wanted to use our smell machine in the entertainment industry, providing a heightened experience to the mainstream but maintaining our goal of helping those with sensory difficulties.

In our first prototype, heat was decided as the simplest way of vaporising our samples of scent. We raised the idea of using old heat coils found in e-cigarettes, supporting the environment. However, after in-depth preliminary research, we found that many of our scents would likely be oil-based, raising safety risks in relation to any heat involved in our system.

Therefore, we were forced to seek out alternatives. After many meetings and suggestions, we decided on vaporisation through vibration, using a nebulizer. Whilst this would be more complicated, it met our safety standards whilst delivering our intended goal.

From there, we started looking for smells. Initially, we wanted to make these ourselves, but in light of health standards and safety regulations for inhaled substances, we decided to opt for certified manufacturers.

Our next decision was whether to create smells from controlled amounts of notes, such as vanilla, lemon, mint etc to achieve specific scents. However, after much research, the fine control of amounts of liquid vaporised proved to be a much more complex task than expected. We investigated use of servos and tubing - where the number of servo rotations was ratioed for each note. Alternatively, using an automated syringe. However, this would take too long to create the scent when it is needed quickly for the consumer at a given moment.

Therefore, we decided to release the scent directly into the air. After discussing with our science teachers, we learnt that they should combine as they diffuse in the air, resulting in the same effect.

It was at this point, when things began to look more promising and achievable. That is why we wanted to challenge ourselves a step further.
Our 4D experience would not only be scent, but now accompanied with wind machines and mist. This would level up our device.

Range of approaches to the project - 1.3 and 1.4

There are several ways in which this project can be tackled in meeting our goals of heightening the experience of the consumer when interacting with entertainment.

We considered producing a research project, which would involve delving deep into the factors that affect the overall stimulation from entertainment. From there, we would find our own trends and any variables we conclude can affect the overall reaction. It could also meet our secondary objectives of helping those with sensory issues better experiencing media as we could research how they differ from the mainstream and perhaps what they can do or film producers can do to blur the line between people's engagement of media.

We discussed this option in a meeting and saw many positives of it, however, the overall effect we could have on our objective seemed minimal. We lack the research ability to collect our own primary data, answering our specific questions, so we would have to continuously use secondary data to make our conclusions. This felt as if it weakened our impact and would mean perhaps less motivation to take on the project as things are not as 'exciting'.

Another approach we considered was starting a blog, documenting things by video and presenting our findings in that way. This would meet our objective in being easily comprehendable by our target audiences, telling them about our findings in entertainment.

However, like the research project, we felt real change is often not from telling someone something, it is from making something. Not only do we think that would meet our objective more strongly, but the idea excited the whole team - a hands-on project meant much more to be learnt. A journey that ensured mistakes but also learning that perhaps a research project would lack.

When we reached this conclusion, we ended the meeting and booked another one, where we would discuss what we would be making.

We started with the idea that we wanted to create a more sensory atmosphere when consuming media. Initial ideas were spatial audio, a sound system which encourages you to place yourself in the world before you. This concept clearly exists already so we seeked out an innovation opportunity for this project. However, we couldn't think of anything directly related to sound.

All of our ideas lead us to the other senses. The strongest being smell.
This is how we settled on our 4DX machine, lead by smell and supported by the experiences of wind and mist.

Over the other options, it has several very clear advantages. Firstly, it means we are doing something our research tells us has been scarcely attempted, far more innovative than using other's research to try and put together ideas, which we would too find much more challenging.

When the idea first came up, we all recognised it was ambitious, but that is what made it so interesting. We wanted to experience the mistakes and develop our skills as a team. It was a project that was able to best use our whole teams skills from coding to engineering - skills that would perhaps have not been used in a research project.

Presentation

Attendance: All

On the 11th February 2026, we presented to the entire STEM club, in front of many students, technicians and teachers. This was an exciting opportunity to share our work with our school. We created a set of slides and presented them, before opening the floor to many questions, which we have recorded below, considering how we will fix any issues raised.

We recognise that many of these concerns appear to be for a product entering the consumer market. Therefore, whilst we identified fixes, we established these would not be necessary for our proof of concept aims.

See our presentation attached alongside this document.

Questions	Fix
How can we ensure the product doesn't malfunction	Utilising detection/ safety components such as fuzes and ensuring measures are taken to prevent electronic issues
How can we implement the emergency shutdown	Through game script Power cut fuze
How can we have an individual from the disabled community test the product	We would email SED centres, asking for the opportunity to display our exciting project.
How will we source the product while ensuring it meets safety standards	Ensure to have sufficient certification from reliable associations
What is the budget that we have	Around £100
How can we make this more user friendly	The goal is not to be user friendly and mostly aimed at functionality

Purchased

Core computer system

- Arduino UNO R3

Scent hardware

- Grove atomizer module
- Foam scent sticks
- 5V mini fan (30–50mm)

Relay switching

- 2-channel or 4-channel 5V relay module (optocoupler-isolated)
- Flyback diodes (optional but recommended)

Wiring & electronics

- Breadboard
- Jumper wires (M-M & M-F)

Mechanical

- Enclosure
- Fan mount
- Silicone tubing

Scents

When looking for scents, we prioritised safety, looking at only sellers who were verified and IFRA-complient. After this, we prioritised scent availability as well as cost.

Decided list:

- Rainforest → Link
- Baked Bread → Link
- Merry Morning/ Christmas →scent: Link
- Ocean Luna → Luna

After much experimentation, using 10ml measuring cylinders and going up in increments of 1ml of scent to water ratio, we discovered that the optimal combination was 4 to 6 parts scent to water.

We could not just have a purely scent mixture as through testing, our diffusers weren't able to provide the same distribution of liquid for the scents as they could with water. This was likely due to the densities of the various liquids.

We experimented until the scent could be smelled from 50cm away from the source within 8 seconds, balancing the factors of strength of smell and distribution for optimal results.

The Game

Video explanation of the process of making the game and the connectivity:

My Movie 2.mov

The game was made using Unity and used a pre-coded character. We then made a platform that was split into four environments, corresponding to different and unique smells. We made the environments by using free assets from the unity store which were limited but we picked the best ones for the environments.

Issues faced:
When making the ocean environment, it was difficult to make a simple model / or find one.
Solution:
Find a sufficient substitute to replicate the environment of the ocean. Therefore I have chosen to use a beach.

Issues faced:
By not having prior experience with Unity, it has caused for a large majority of the time being spent on learning the functionality
Solution: Unfortunately like a lot of things there are not short cuts, and only get better by practising. For this case I was able to accelerate the pace that I learnt through the use of youtube videos listed below.

https://www.youtube.com/watch?v=3wcMtJK8iIY
https://www.youtube.com/results?search_query=how+to+import+asset+into+unity
https://www.youtube.com/watch?v=FiW69KIsQz0

Issue faced:
Incompatibility issues with computer due to lack of processing power
Solution: I was able to borrow my brother's laptop to complete the work.
Learnt: With many projects we must stop relying heavily on a singular devise, as well as constructing backup plans and thinking far ahead to ensure that major blockage can be prevented

Building the bridge between the game and the usb:

The sender - Unity:

When an event occurs within the game, such as the character crossing into an invisible trigger box a script is triggered. The script is unity using C# script to send a single character such a “1” through to the usb

The bridge

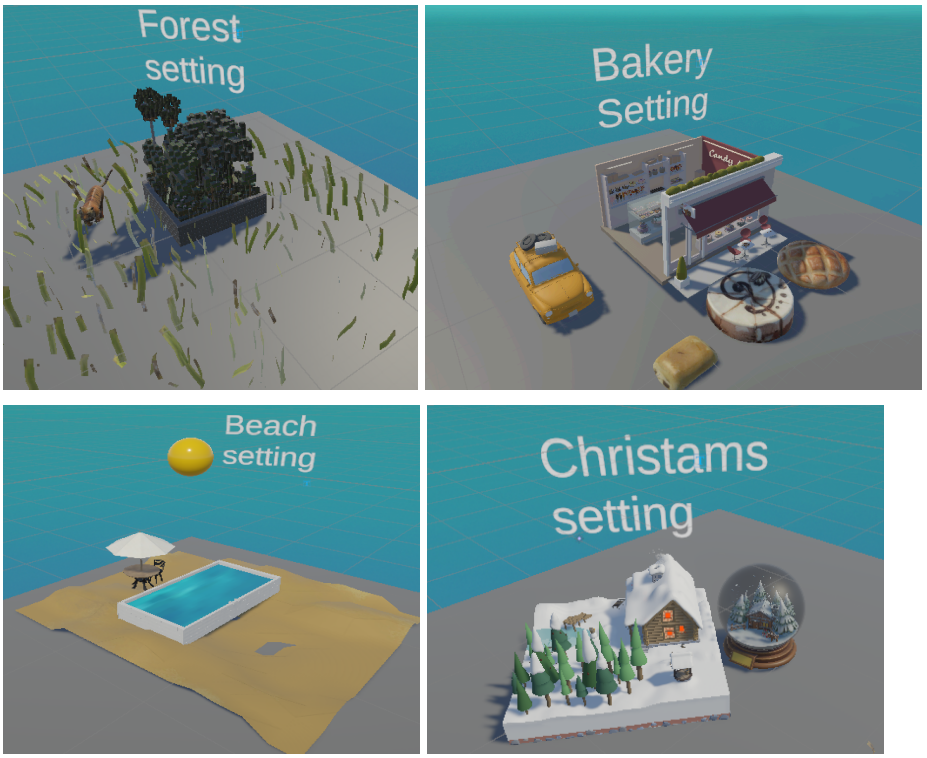
The USB cable is the direct communication pipelines which is configured to run to specific ports and speed which is measured in braud rate allowing for effective communication between unity and arduino.

To ensure that this occurs error free this is a one way channel from Unity to Arduino

The Listener (arduino)

The arduino is at a constant listen mode where it is open to receiving any data from the usb cable. As soon as data is in the form of characters as “1” or “2”, it triggers the if statement in the code to send power to the atomizer.

Images from the game environments:



The links to the components of each are below:

Forest setting:

- *Grass* - Link
- *Tiger* - Link
- *Tree* - Link

Bakery setting:

- *Bakery* - Link
- *Pastries* - Link
- *Car* - Link

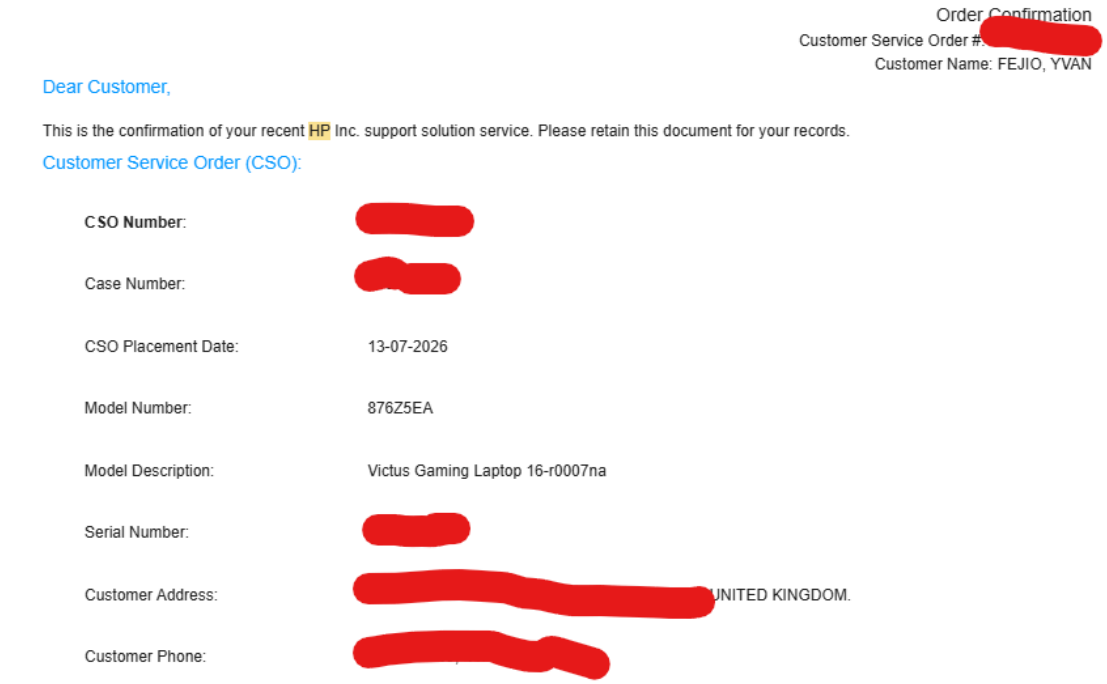
Beach setting:

- *Sand* - Link
- *Pool* - Link

- *Table set* - Link
- *Sun* - Link

Christmas setting:
- *Hut* - Link
- *Snowglobe* - Link

Evidence of technical issues:



Order Confirmation
Customer Service Order #: [redacted]
Customer Name: FEJIO, YVAN

Dear Customer,

This is the confirmation of your recent HP Inc. support solution service. Please retain this document for your records.

Customer Service Order (CSO):

CSO Number:	[redacted]
Case Number:	[redacted]
CSO Placement Date:	13-07-2026
Model Number:	876Z5EA
Model Description:	Victus Gaming Laptop 16-r0007na
Serial Number:	[redacted]
Customer Address:	[redacted] UNITED KINGDOM.
Customer Phone:	[redacted]

Hardware for Myrodia project

My original idea was to use MQTT to communicate wirelessly between a PC running a Unity game and a microprocessor-controlled ultrasonic atomizer.

Ultrasonic Atomizer

Option 1

Seek a 3.3v ultrasonic atomizer module that can be used with a Raspberry Pi.

I found the Keyestudio Atomization Modtule which says it can use 3v to 5v. However, this seems to be unobtainable now. I haven't been able to find any other 3.3v ultrasonic atomizer modules.

Option 2

You can buy cheap 5v USB-powered ultrasonic atomizer modules that could be controlled by relay but it would be a clunky solution and require a lot of fiddly modifications to make it work as you'd have to hack either the USB cable or the USB socket on the board. Additionally, the boards I have seen have a push button that needs to be pressed, so you would have to bridge the switch by soldering a wire across it.

Option 3

The simplest solution I can find is the Grove Atomizer Module, which is available to buy in the UK for around £10.

Grove boards are a way of interconnecting electronic devices to a wide range of microprocessor boards (e.g. Raspberry Pi, Arduino, ESP32 etc.) by means of a Grove base board that fits over the pins of the microprocessor board. Grove base boards cost around £10 (depending on which) and you would need only one. The Grove boards take care of driving the devices correctly, so as not to damage the components or the microprocessor.

You could buy four of these Grove Atomizer Modules and connect them onto a Grove base. The Grove base would also allow additional components like a fan to be connected.

The problem is that the Grove Atomizer Module operates at 5v and would damage a Raspberry Pi as the Raspberry Pi uses 3.3v logic level. You could get round this problem by using different microprocessor board that can use 5v logic level, such as an Arduino Uno.

This then causes another problem, which is that the basic Arduino Uno has no network capability. An Arduino with WiFi (Uno R4 WiFi) costs around £25.

An alternative solution would be to use the Grove modules but use a basic Arduino Uno (with separate power supply) and connect it to the PC using serial commands over USB . The separate power supply is needed because you'll be powering four/five devices via the attached Grove shield. In principle, the Arduino could be controlled via serial commands sent direct from Unity using C#. This is not as neat as using MQTT over WiFi, but would be a lot

simpler. This would be my recommendation for a “proof of concept”. It just doesn’t seem possible to find ultrasonic atomizer modules that will work with a Raspberry Pi.

Shopping list

- Arduino Uno [I think you said you had one] plus barrel jack power supply [Do you have one?]
- Arduino Grove Base Shield £2.56 plus VAT
- Grove Atomizer Module x 4 - £31.96 plus VAT (£7.99 plus VAT each)
- To waft the scent – a Grove mini-fan

If you buy the above from Farnell as per the links, it’ll all cost £52.91 including free delivery (not including Arduino + power supply).

Notes

1. The other thing to note is that you need to use C++ on the Arduino, rather than the Python you would have probably used on the Raspberry Pi.
2. I found this video which will give you some ideas for coding both in Unity and on the Arduino.
3. Seeed Wiki
   https://wiki.seeedstudio.com/Grove-Mini_Fan/#features
   https://wiki.seeedstudio.com/Grove-Water_Atomization/

Steve Smythe
18/03/2026

PS - You’ll need to explain in your CREST report why you decided to switch from a Raspberry Pi to an Arduino solution - after discussion with your mentor and per my rationale above.

Conclusion

To have a meaningful conclusion, we wanted to compare our completed project with the objectives we set out for ourselves before beginning.

The primary objective was to create a device that connected to a game and in real time, automatically responded to location-induced signals to release smells and humidity. This was to create a “4D” like experience.

To judge our success in doing so, we looked at our individual aims as members of a team and our personal judgements on our work as well as those of others, inside and outside of the group.

Smell dispensation

Releasing the smells was definitely the most difficult part of this project, as our use purpose required quite a specific type of technology. This was primarily due to the fact that unlike other scent devices, ours needed to quickly start and stop release, where smells didn’t linger post-stimulation. However, by setting this criteria, many of the options available raised safety concerns, challenging another one of our aims, which was ensuring safety.

Furthermore, this was the aim where we were initially *too ambitious*, as we envisioned mixing scents to create “custom” smells for a very large range of in-game sites. However, through planning and many meetings, this design seemed far too complicated. Ideas involved hydraulics, syringes, motors and more. It was certainly very exciting, however the main problem was that it would mean increased dispensation time. We couldn’t meet our aim of having an increased user experience if they would smell the scent over 5 seconds after the stimulation. Another problem for this complex design would be increased failure points, which any engineering project should aim to minimise. It was an example of where the simplest solutions are often the most effective. Hence we decided not to mix scents and release them directly, with several points of dispensation, one for each scent, rather than an overall site where scents would be ‘injected’ into.

From there, we had another obstacle of how the scent would actually be dispensed. The fastest would be heat - the most obvious pick. However, returning to our aim of safety, this was not viable. Not only did we not want the risk of any potential fire hazards, but our choice of scent would be heavily reduced using heat as there are some serious safety concerns regarding respiratory protection and ignition. That is why in the end, we decided on vibration, the perfect balance between speed of response and safety.

Overall, there was a huge amount of research into this and it leaves many exciting opportunities in the future to increase the complexity of this project, as we continue to learn. We hope to come together on this again when we have more time and experience.

Mist and Vortex aspects

This was considered an ‘addition’ to our project, complimenting the smell base. We experimented with both and noticed the mist was very successful. We worried in our planning phase that it would be warmed by the vibration and not have that ‘refreshing’ mist effect, however that was not the case. It was just as intended and really elevated the project. However, the vortexes, whilst planned extensively, were very expensive to get the level of power we wanted. We experimented with standard fans and they worked but they just didn’t give the user experience we were looking for. The initiation of the fan wasn’t as sudden of a gust as we had wished, so in the end we decided this is a part of the project to leave out for now. If we were to improve the project at a time where we had greater resources, this would certainly be one of the first things we went back to.

Game development

The game was developed using Unity, and used a pre-coded character where we designed four different environments all with different smells, so we could showcase the breadth of smells Myrodia could present. We included a beach, a forest, a bakery, and a cabin. This was done with limited programming because we wanted our focus to be on the Myrodia device instead. We made these environments by using different assets from the unity store but this proved to be more difficult than imagined because using unity was more complicated than just drag and dropping assets, but we learnt how to navigate the different menus and access different assets from the unity store.

Unity proved to be a harder task than we first thought but we learnt a lot in the process and it is a good skill for us to have. We will surely be using unity again in the future and could use coding within unity to create a more complicated game next time

Linking everything together

The device that we made was used with an arduino, where we used C++ and it released smells depending on the environment the character in the game was in. The programming was very complicated, so we used youtube videos and our previous experience with Arduinos to aid us with this. It was a very fun part of the project to work on and involved a lot of problem solving and error spotting. The arduino was then linked to four different vibrating devices, which connected to different smells, and the different devices will turn on depending on which smell needs to be released.

We then had to link the game with the Myrodia device which was a difficult task to do. We decided in the end to link the two using a wired connection as this proved to be the easiest. The other option was to create a LAN network and connect it wirelessly but this would have been a lot more complicated.

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